



Francisco Javier Guinot Almenar

Software Engineer | Game Engine Developer | Low-Level Programmer

✉ whitewholffox@gmail.com ☎ [642187996](tel:642187996) 📍 Valencia, SPAIN 🌐 [Portfolio](#)
🌐 [Linkedin](#) 🐙 [Github](#)

Skills

📁 Languages

C, C++, C#, Python, SQL, GLSL, Bash /
Shell Scripting, Batch Scripting, x86/x64 Assembly
(Reading/Debugging)

🔧 Computer architecture

Cache-friendly programming, Manual memory
coherency & management, Data alignment (SIMD),
Cache miss reduction, Thread pooling, Task scheduling,
Memory alignment (16-byte boundary profiling), GPU
Link-list management

🔗 Graphics & Low-Level

Vulkan, OpenGL, Multi-threading (Lock-free primitives),
Engine Development, Custom memory allocators,
Real-time simulation, Deferred shading, Volumetric
rendering, Shadow mapping (Texture arrays),
GPGPU (Compute shaders), SPH (Smoothed particle
hydrodynamics)

🔧 Hardware & systems

Peripheral communication protocols (USB / Industrial
protocols), Real-time operating systems (RTOS)
concepts, Linux/Windows internals

📁 Tools & Frameworks

Git, CMake, Visual Studio, Unity, Unreal Engine 5,
Jira, Subversion (SVN), RenderDoc, NVIDIA NSight
(Profiling), STB Image, OpenAL

Languages

Spanish

Native

English

Full professional proficiency

Interests

Low-level hardware architecture

Deep interest in understanding CPU/GPU internals -
cache hierarchy and how silicon executes code

Retro-computing & emulation

Developing and studying emulators to understand how
legacy hardware managed limited resources through
extreme optimization

Mathematics for compute graphics

Constant learning of linear algebra - calculus and
numerical methods applied to physics and rendering,
Kernel calculus (Spiky/Poly6), Numerical Integration

Physics simulations

Exploring GPGPU techniques to simulate complex
natural phenomena (fluids - soft bodies and
thermodynamics) in real-time, Spatial Partitioning
(Voxel/Grid)

Summary

Performance-driven software engineer specializing in high-fidelity simulation and graphics engine development. Expert in modern C++, manual memory management, and hardware-level optimization. Extensive experience building graphics engines from scratch (Vulkan/OpenGL) and real-time simulators. Focussed on Data-Oriented-Design (DOD) and high-efficiency architectures (ECS) to maximize CPU/GPU throughput and ensure deterministic, low-latency systems.

Technical Philosophy

I advocated for minimal-dependency development to maintain total control over the software lifecycle. I prioritize "hardware-aware" code, understanding that every software decision from smart vs. raw pointers to data layout in memory directly impacts the latency and efficiency of the final system.

Experience

Simmodar Training Solutions (LSyM - University of Valencia) Burjassot, Valencia
Software Developer / Systems Programmer Oct 2024 - Present

- Developed and optimized high-performance backends for professional-grade simulators (Heavy machinery, maritime and logistic systems).
- Engineered core simulation logic and real-time physics using C++ and PhysX5 with a strict focus on low latency response and physical accuracy.
- Hardware-in-the-loop (HIL): Developed communication layers to synchronize physical cabins and sensors with the simulation engine.
- Optimized rendering pipelines to ensure stable frame rates in high-complexity environments, managing resources and memory to prevent bottlenecks
- Utilized professional version control and project management workflows (SVN / Jira / Mantis)

[LSyM Web](#)

Projects

Custom C++ OpenGL Engine (ECS, Deferred & Volumetric Pipeline)

Sep 2023 - Jun 2024

- **Core Architecture:** Developed a modular 3D engine from scratch using **C++17** and **OpenGL 4.6**, implementing a Data-Oriented **Entity Component System (ECS)** to maximize cache hit rates via contiguous memory buffers.
- **Rendering Pipeline:** Engineered a hybrid **Deferred/Forward** renderer featuring advanced lighting kernels, including **Volumetric Lighting (Raymarching)** and PCF Shadow Mapping for Directional, Spot, and Point lights using **Texture Arrays**.
- **Asynchronous Systems:** Designed a custom **Thread Pool (Task Manager)** to offload heavy computations and implement asynchronous mesh and texture loading, ensuring a stutter-free main loop.
- **Tooling & Internals:** Implemented a **GPU-based entity picking system** via color-buffer ID encoding and a custom Linear Algebra library optimized for 3D transformations and Frustum calculations.
- **Asset Management:** Integrated a 3D spatial audio core (OpenAL) and a custom binary asset bundler with **XOR encryption** based on PGM key-image generation.
- [GitHub project \(Public\)](#)

Vulkan Graphics Engine (Cross-Platform)

(Ongoing)

- Ongoing development of a graphics engine designed for maximum hardware control (Windows / Linux / Android)
- Manual management of Vulkan memory allocator, synchronization primitives (fences / semaphores), and command queues.

High-Performance GPU Cellular Automata Framework

Sep 2023 - Jun 2024

- Architected a GPGPU simulation suite for complex automata (Lenia, SmoothLife, Conway) using **OpenGL Compute Shaders** on **1024x1024 grids**.
- **Performance Metrics:** Achieved execution times of **200µs** for discrete kernels (Conway/SmoothLife) and **6ms** for continuous convolution-based kernels (Lenia), ensuring high-frequency simulation within strict frame budgets.
- **Optimization:** Engineered an optimized Lenia version using **SSBOs (Shader Storage Buffer Objects)** and multi-pass kernels to decouple neighbor sampling, significantly reducing redundant texture fetches.
- **Memory & Sync:** Managed data integrity using "ping-pong" texture buffering and **explicit memory barriers** (glMemoryBarrier) to prevent race conditions during asynchronous GPU state updates.
- **Mathematical Kernels:** Implemented continuous Gaussian Bell kernels and spatial pre-computation for neighbor indexing, maximizing throughput of the GPU's SIMD architecture.

SIMD-Optimized Metaballs (x86/x64 Assembly)

Sep 2022 - Dec 2022

- Authored a real-time metaballs visualizer using manual register allocation and SSE instructions to achieve high-performance pixel-per-pixel calculations on legacy SDL systems.

Massive-Scale GPU Fluid Solver (SPH & Custom Spatial Partitioning)

Sep 2023 - Jun 2024

- **Core Physics:** Engineered a high-performance **Smoothed Particle Hydrodynamics (SPH)** solver capable of simulating **400k+ particles at 40 FPS** in real-time.
- **Custom Architecture:** Developed a novel **Nested Voxel-Cell spatial partitioning** system to solve the neighbor search problem on the GPU, utilizing a custom-built "Remapping" protocol for dynamic list management in **SSBOs**.
- **GPGPU Pipeline:** Implemented a multi-stage **Compute Shader pipeline** (Integration, Spatial Binning, Density Accumulation, Force Application) with strict memory alignment and **GPU synchronization primitives**.
- **Optimization:** Achieved massive throughput by minimizing atomic operations and leveraging a localized grid-based search, significantly reducing the $O(n^2)$ complexity of particle interactions.
- **Technical Challenges:** Managed high-precision floating-point synchronization and data integrity across massive GPU buffers, handling the inherent complexity of concurrent simulation kernels.

Education

ESAT (Escuela Superior de Arte y Tecnología)

HND in Computing and Systems Development

Game Programming

Valencia • Sep 2021 - Jun 2024

- Intensive focus on **Advanced C++**, Data Structures, Computer Architecture and Gaming engines (Unity / Unreal Engine 5)
- Developed custom game engines from scratch using C++ and Graphics APIs (OpenGL).
- Gained a deep understanding of memory management, optimization, and software engineering patterns applied to real-time systems.

[E.S.A.T Web](#)